

Ocean West

Austin, TX | oceanw2003@gmail.com | (210) 632-7142 | linkedin.com/in/oceanw2003 | US Citizen

EXPERIENCE

Composites Materials and Processes Integration Lead - Combustion Team

May 2025 – Present

Longhorn Racing, UT FSAE

Austin, TX

- Coordinated projects and introduced a design documentation system for carbon fiber steering wheel, pedal-box composites integration, sandwich panel structures, body panels, and tube assemblies in compliance with ASME, ISO, and ASTM standards.
- Constructed and maintained a comprehensive library of composite material properties within ANSYS ACP, which equipped team members with the resources to fix the biggest causes of crashes.
- Optimized the carbon fiber steering wheel design through FEA and layup adjustments, achieving a **50% weight reduction** from the previous year with validated stiffness and strength retention.
- Performed tensile, 3-point bend, epoxies/adhesive tests, and torsional tests on composite specimens to validate ANSYS simulations and characterize material performance; led to **FOS ranging between 1.5 and 8**.
- Implemented advanced manufacturing methods including Compression Molding, Resin Infusion, Sheet Molding Compound (SMC) for forged carbon fiber, CNC machining, and high-temp 3D-printed tooling.
- Refined advanced carbon fiber layup techniques in prototype development, enabling the team to manufacture complex geometries with enhanced precision; led to a **10% reduction in material waste**.
- Orchestrated comprehensive recruitment strategy that onboarded six engineers from a pool of over 100 applicants, achieving a **15% improvement** in first-time build success rate.
- Created Parametric CAD models for composite structures **enhancing efficiency by up to 60%**.

Composites Engineer - Combustion Team

Sep 2024 – May 2025

Longhorn Racing, UT FSAE

Austin, TX

- Engineered and fabricated carbon fiber parts (airfoils, panels, muffler, bodywork) to optimize performance and manufacturability, contributing to the team's **11th-place finish** in design.
- Developed an Excel-based calculator integrating CAD surface densities and material properties to automate composite weight estimation, accelerating design iterations **by 15%** and improving structural performance of manufactured parts.
- Applied 3D printing for rapid prototyping and mold fabrication, improving turnaround on a new front wing design, side pods and steering wheel prototypes.

Undergraduate R&D Intern - Fatigue Testing of 3D Printed Components

Jan 2024 – Aug 2024

Margie and Bill Klesse College of Engineering and Integrated Design, UTSA

San Antonio, TX

- Conducted **over 100** axial fatigue tests on additively manufactured titanium components using an MTS Landmark to evaluate material performance under cyclic loading.
- Processed and analyzed experimental fatigue data to assess mechanical behavior and microstructural influences in additively manufactured metals, calculating Equivalent Initial Damage Size value (EIDS) for Ti-6Al-4V samples.

PROJECTS

RC Car Project: MF-5 | GD&T CAD/CAM, Iterative Design, Fabrication, Vehicle Dynamics

Jan 2025 – May 2025

- Secured **#1 qualifying position** out of 40 teams with the fastest lap time (18.46s over three laps).
- Developed a custom steering system with optimized Ackermann geometry, implementing 1° negative camber and a 7° steering angle improvement that enhanced cornering efficiency and secured the top design score.
- Conceptualized an enhanced layout for internal components, enabling a **7% denser** arrangement while upholding cooling efficiency, ensuring the vehicle's optimal performance under race conditions, and maintained structural integrity.

EDUCATION

University of Texas at Austin

Aug 2024 – May 2027

BS Mechanical Engineering | GPA: 3.51/4.0

Austin, TX

- Affiliations:** Longhorn Racing Combustion, ASME
- Relevant Coursework:** Structural Analysis, Composite Laminate Theory, Mechanics of Materials, Fluid Mechanics, Materials Engineering, Machine Elements, Mechatronics, Heat Transfer, Dynamic Systems and Controls, Materials Processing

University of Texas at San Antonio

Aug 2022 – May 2024

BS Mechanical Engineering | GPA: 3.89/4.0

San Antonio, TX

- Affiliations:** Undergraduate Research Intern, AIAA, Bold Scholars UTSA
- Relevant Coursework:** Dynamics, Thermodynamics, Thermal Fluid Systems, Numerical Methods, Differential Equations

SKILLS & ABILITIES

- Software & Tools:** SolidWorks, ANSYS Mechanical, Matlab, Python, Arduino, Bild PDM, Orca Slicer, HyperMesh
- Manufacturing:** 3D Printing (FDM), Prepreg Layup, Fiber Orientation Optimization, Core Material Integration, Surface Finish Techniques, Tooling Design, Material Testing
- Analysis & Testing:** Digital Microscopy, Wind Tunnel Testing, Data Acquisition, Signal Processing, Non-Destructive Testing

ADDITIONALS

- Awards:** FSAE International Competition **11th Place** in Design (2025), President's List (1x), Dean's List (2x), Sapientia Award
- Activities:** Golf, Digital Photographer, Astronomer, Motorsport & F1 Enthusiast

Fun Fact: I have never had a Red Bull energy drink.